

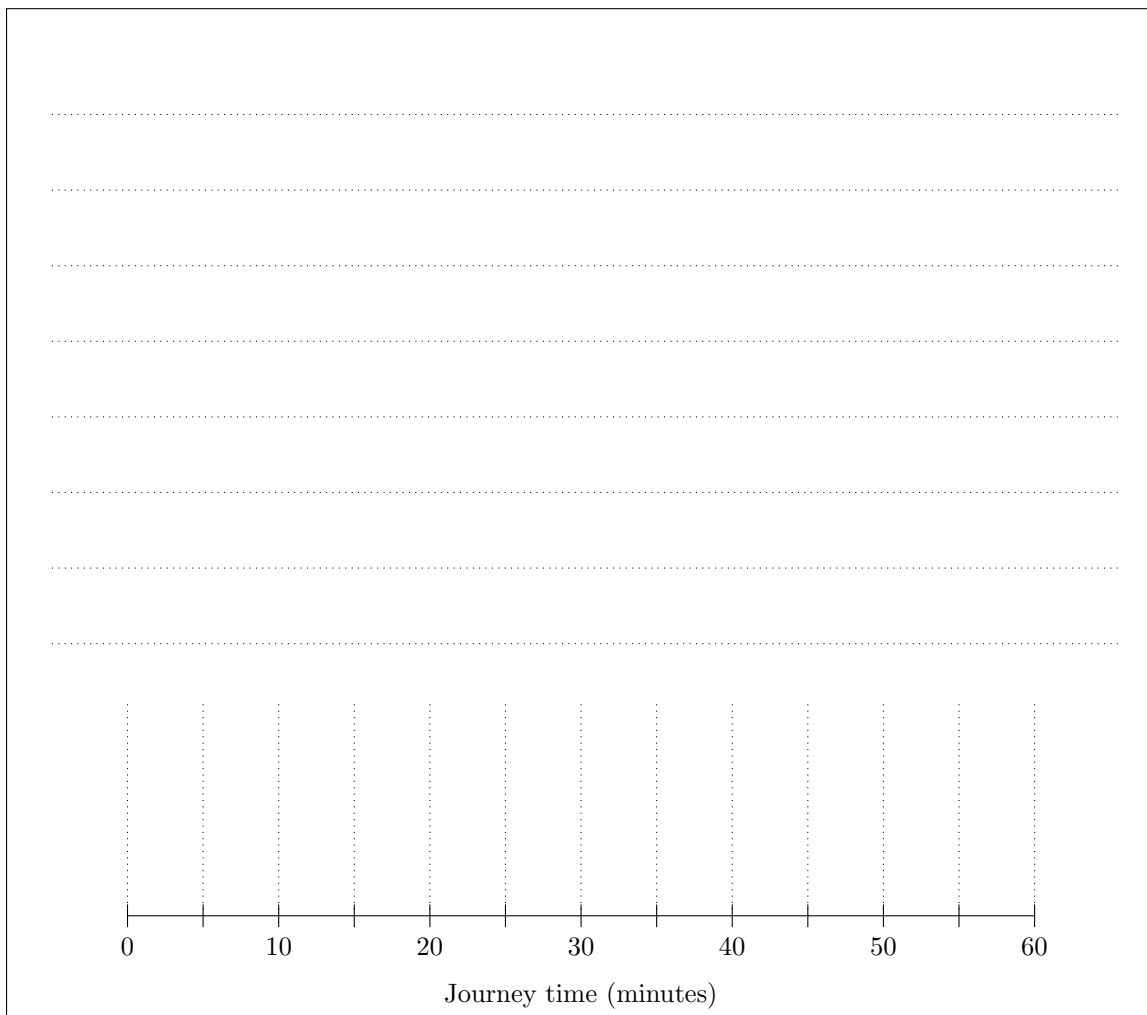
1.5 Classwork: Box-and-Whisker Plots

Use your GDC. Give answers exactly, or correct to three significant figures. An outlier is a value more than $1.5 \times IQR$ below Q_1 or more than $1.5 \times IQR$ above Q_3 . On a box-and-whisker plot, mark an outlier with a cross ($\times$).

1. Fifteen students record the time, in minutes, of their journey to school.

24, 18, 31, 12, 58, 22, 26, 15, 30, 21, 35, 20, 28, 33, 25

- (a) Write down the median, the lower quartile Q_1 and the upper quartile Q_3 .
- (b) Find the interquartile range.
- (c) Show that 58 is an outlier.
- (d) On the scale at the bottom of the box, draw a box-and-whisker plot of the journey times. Mark the outlier with a cross.



Nickname:

2. The box-and-whisker plot at the bottom of the box shows the test scores of the students in Class A. The scores of the students in Class B are summarized in the table.

Class B	Minimum	Q_1	Median	Q_3	Maximum
Score	22	56	66	76	94

- (a) Write down the median and find the interquartile range of the Class A scores.
- (b) Show that the lowest Class B score is an outlier, and that the highest Class B score is not.
- (c) The lowest Class B score that is not an outlier is 41. On the same scale as Class A, draw a box-and-whisker plot of the Class B scores. Mark the outlier with a cross.
- (d) Compare the scores of the two classes. Make one comparison of the centre and one comparison of the spread, quoting values.

